Ali Akarma

AI Researcher | Agentic AI, Trustworthy Machine Learning
Islamic University of Madinah, Saudi Arabia
Email | Google Scholar | GitHub

PROFILE

AI researcher focused on trustworthy and safety-aligned agentic systems, with peer-reviewed publications spanning AI architectures, governance, and real-world deployment contexts. Experienced in designing constrained agent pipelines, analyzing failure modes, and studying robustness in high-stakes environments. Preparing for graduate research in reliable machine learning systems.

RESEARCH INTERESTS

- Agentic AI and autonomous decision-making systems
- Trustworthy machine learning and AI safety
- Robustness and failure analysis of LLM-based systems
- Human-centered AI and accessibility technologies
- AI governance and responsible deployment

EDUCATION

B.S. Information Technology (2023–2027)
Islamic University of Madinah, Saudi Arabia

- Merit-based fully funded scholarship
- Undergraduate researcher in agentic AI systems and AI governance

RESEARCH EXPERIENCE

Undergraduate AI Researcher
Islamic University of Madinah | *Jan 2025 – Present*

- Designed agentic AI systems operating under explicit safety constraints, including restricted action spaces and policy-aligned decision rules.
- Conducted failure-mode and misuse analyses for autonomous systems, focusing on adversarial inputs, unintended behaviors, and oversight gaps.
- Explored LLM-based components within agentic pipelines, including prompt design, orchestration, and robustness considerations.
- Investigated governance mechanisms for preventing unsafe behaviors triggered by untrusted or adversarial data sources.
- Analyzed system trustworthiness in scenarios analogous to log analysis, automated triage, and decision support pipelines.

PUBLICATIONS & MANUSCRIPTS

Peer-Reviewed Conference Papers

- Jan, S., **Akarma, A.**, et al. Blockchain-Monitored Agentic AI Architecture for Trusted Perception–Reasoning–Action Pipelines. IEEE ICCA, 2025.
- Toqeer Ali Syed, **Akarma, A.**, et al. FinAgent: Agentic AI for Personal Finance & Nutrition. IEEE ICCA, 2025.
- Toqeer Ali Syed, **Akarma, A.**, et al. Agentic AI for Cloudburst Prediction & Response. ICBDT, 2025.
- Toqeer Ali Syed, **Akarma, A.**, et al. Agentic AI for Smart Inventory Replenishment. ICBDT, 2025.
- Salman Jan, **Akarma, A.**, et al. Agentic AI for Disabilities & Neurodivergence. ICBDT, 2025.

Accepted / In Press

- Agentic AI for Inclusive Assistive Ecosystems: Architecture, Governance, and Personalized Support for People with Disabilities *(Book Chapter)* accepted, **IGI Global**
- An Agentic Digital-Twin Framework for Customer Churn Prediction and Retention in Retail Banking *(Book Chapter)* — accepted, **Springer Nature**
- FinNutriAgent (FNA): An Agentic AI for Budget and Nutrition Planning — accepted, **ETASR**

Under Review

- Ethical Governance of Agentic AI in Higher Education — under review, ***Heliyon***
- Ethical AI Governance for Cybersecurity in Renewable Energy IoT — under review, ***PLOS ONE***
- Agentic AI-Enhanced Digital Twins for Smart Cities — under review, ***PLOS ONE***
- Use of AI Tools: Guidelines to Maintain Academic Integrity — under review, ***Springer***
- Risk-Aware Multi-Agent Deep Reinforcement Learning for Cloudburst Prediction and Coordinated Disaster Response — under review, **MDPI Mathematics**

SKILLS

Programming & Data

- Python
- Java
- SQL

Machine Learning & AI

- LLM orchestration & prompt design
- Agentic AI architectures
- Human-in-the-loop oversight
- AI safety and alignment concepts
- Preference-aligned AI concepts

Research & Engineering Tools

- LaTeX
- Mendeley
- Experimental design & simulation

Systems & Platforms

- Linux (basic)
- Git (version control)
- API-based integration

CERTIFICATIONS & AWARDS

- Merit-Based Fully Funded Bachelor's Scholarship — Islamic University of Madinah
- Certificate of Appreciation — ICBDT 2025
- Mindware: Critical Thinking — University of Michigan
- Computational Thinking — University of Michigan

Service & Volunteering

Volunteer — Masjid Al Nabawi, Madinah
2024; 2025 (1 month each)

- Supported public-service operations in a large, high-traffic institutional environment.
- Assisted visitors and contributed to orderly, inclusive, and reliable service delivery.